

A SEMI-SUPERVISED METHOD FOR SAR TARGET DISCRIMINATION BASED ON CO-TRAINING

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ABSTRACT

In synthetic aperture radar (SAR) target discrimination, supervised methods, which need lots of labeled samples, are widely used. However, obtaining sufficient labeled samples is usually costly and laborious. In this paper, we propose a semi-supervised SAR target discrimination method based on co-training, which exploits both labeled and unlabeled samples. Firstly, we extract Lincoln features from the training samples and separate them into two subsets according to their physical meaning. Then, the co-training algorithm is utilized to iteratively train two support vector machine (SVM) classifiers on the extracted two feature subsets. Finally, the trained classifiers are exploited to classify the test data. The proposed method is evaluated with the miniSAR real dataset. Experimental results demonstrate that the proposed method is not only superior to the supervised methods with little labeled samples, but also the self-learning which only uses one feature subset.

Index Terms—synthetic aperture radar (SAR), target discrimination, semi-supervised learning, co-training

1. INTRODUCTION

Due to its ability to form high resolution images with relative invariance to weather and lighting conditions, synthetic aperture radar (SAR) has been successfully applied to civil and military applications. An SAR image contains both shape and scattering information, thereby offering a good representation of targets. Therefore, automatic target recognition (ATR) of SAR images has attracted much more attentions during the past decades since it was proposed in 1993 [1][2]. A typical SAR ATR system often consists of three stages [3][4]: detection, discrimination, and classification/recognition. Target discrimination aims at reducing classification/recognition cost by removing both natural and artificial clutter chips obtained in the detection stage. Moreover, as discrimination is the stage linking detection and classification/recognition, it directly influences the performance of the whole system [5][6]. As a result, SAR target discrimination has gained increasing attention during the last two decades.

The existing SAR target discrimination methods mainly focus on the supervised paradigm where all the training data are labeled. Despite the successfulness of the supervised methods, a large number of labeled samples are needed to obtain good performance. However, obtaining sufficient labeled samples is always time consuming and high-cost. Moreover, since the available training samples are limited in SAR targets discrimination, the underlying probability distribution function of the image may not be properly captured and the risk of overfitting problem still exists. These limitations of the supervised learning foster the development of semi-supervised learning methods.

Semi-supervised learning refers to a general class of machine learning techniques that make use of both labeled and unlabeled data for training, typically a small amount of labeled data and a large amount of unlabeled data [7]. Existing semi-supervised learning approaches include: self-training, co-training, semi-supervised support vector machine (SVM), and graph-based semi-supervised learning, etc. Co-training is a well-known semi-supervised learning algorithm which was proposed by Blum and Mitchell [8] in 1998. It is usually applied to datasets that have a natural separation of their features into two disjoint sets, which are regarded as two views of the data. At each iteration of co-training progress, two learners are trained independently from the two views and required to label some unlabeled examples for each other to argument the training set. The iterative process goes until some stopping criterion reaches. The successfulness of co-training lies in utilizing the large amount of unlabeled samples and the information from the other view. It is observed that over 50 features have been presented for SAR target discrimination during the last 20 years [9]. However, to our knowledge, the co-training method has not yet been applied in SAR target discrimination.

In this paper, a target discrimination method based on the semi-supervised co-training is proposed for the SAR images. In the training stage, firstly, six Lincoln features are extracted from the unlabeled samples. We randomly pick out a few of samples and label them by hand. These manually labeled samples are treated as the initial training data sets. Then the Lincoln features are split into two sets according to their physical meaning and used to train two SVMs corresponding to the two sets of features via co-training algorithm. In the test stage, six Lincoln features are firstly extracted from the test data and divided into two sets

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same as the training stage. Then the trained SVMs are applied to discriminate the test data.

The remainder of this paper is organized as follows. Section 2 presents our proposed semi-supervised SAR image target discrimination method. Section 3 analyzes the experimental results on the real SAR data. Section 4 concludes this paper.

2. PROPOSED METHOD

The proposed SAR target discrimination method consists of the training stage and test stage, as shown in Fig. 1. Meanwhile, the training stage mainly comprises two parts: 1) the extraction of Lincoln features; 2) semi-supervised co-training on Lincoln features. In the test stage, after the two corresponding feature sets are extracted from the test samples, the two trained optimal SVM classifiers are incorporated to discriminate the test samples. Samples that are classified as the targets by both two SVM classifiers are finally regarded as the targets, otherwise samples are regarded as the clutter.

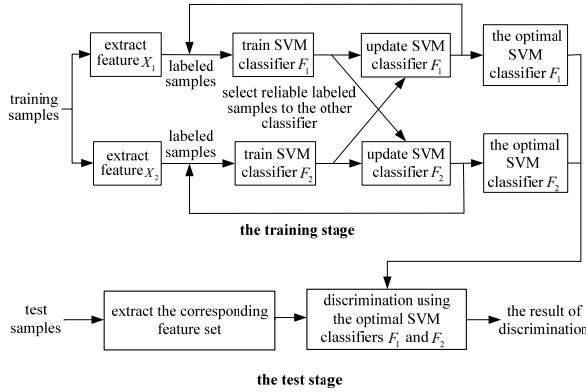


Fig. 1 The flowchart of proposed method.

2.1. Lincoln Features Extraction

Lincoln features are designed by the Massachusetts Institute of Technology (MIT) Lincoln Lab and widely used in the target discrimination [6]. In our semi-supervised target discrimination method, we choose six Lincoln features, including three features related to the texture characteristic and three features related to the contrast. The texture-based and contrast-based features are specially designed for SAR target discrimination and describe the sample characteristics from two different views.

2.1.1. The texture-based features

The texture-based features, containing log standard deviation, the fractal dimension and the weighted-rank fill ratio, are proposed by Burl *et al* [10]. from Lincoln Laboratory.

1) *The log standard deviation feature*: The standard deviation of a chip is a statistical measurement of the fluctuation of pixel amplitudes and can be calculated as follows:

$$\sigma = \sqrt{\frac{S_2 - S_1^2/N}{N-1}} \quad (1)$$

$$S_1 = \sum_{x,y \in \text{region}} 10 \log_{10} p(x,y)$$

$$S_2 = \sum_{x,y \in \text{region}} [10 \log_{10} p(x,y)]^2$$

$p(x,y)$ represents the amplitude value of the pixel located at (x,y) , and N is the number of pixels in the region.

2) *The fractal dimension feature*: The fractal dimension represents the spatial dimension of a chip and can measure the spatial distribution density of the bright pixels in a chip. Since the bright pixels of a target chip are more densely distributed than those of clutter, the fractal dimension can be utilized to discriminate the target from clutter. The fractal dimension is defined as the Hausdorff dimension of K brightest pixels in the chip

$$H_d = \frac{\log_{10} N_1 - \log_{10} N_2}{\log_{10} d_2 - \log_{10} d_1} \quad (2)$$

where N_1 is the number of 1-pixel-by-1-pixel boxes needed to cover all K pixels and N_2 is the number of 2-pixel-by-2-pixel boxes needed to cover all K pixels; d_1 and d_2 denote the size of boxes, and $d_1 = 1$, $d_2 = 2$. Usually, the fractal dimension of clutter is less than 1, while the fractal dimension of a target is between 1 and 2.

3) *The ranked fill ratio feature*: This textural feature measures the percentage of the total energy contained in the brightest scatters of a chip. It is defined as follows:

$$\eta = \frac{\sum_{k \text{ brightest pixels}} p(x,y)}{\sum_{\text{all pixels}} p(x,y)} \quad (3)$$

2.1.2. The contrast-based features

The contrast-based features consist of the maximum constant false alarm rate (CFAR), mean CFAR and percent CFAR [3]. The CFAR algorithm is firstly used to create a CFAR image and the contrast-based features can be obtained. The detector can be described by

$$\frac{p_t(x,y) - \hat{\mu}_c}{\hat{\sigma}_c} > K_{CFAR} \Rightarrow \text{target},$$

$$\frac{p_t(x,y) - \hat{\mu}_c}{\hat{\sigma}_c} \leq K_{CFAR} \Rightarrow \text{clutter}, \quad (4)$$

where $p_t(x,y)$ represents the amplitude value of test pixel, $\hat{\mu}_c$ and $\hat{\sigma}_c$ are estimates of the local mean and local standard deviation, respectively, of the surrounding clutter,

and K_{CFAR} is a constant threshold value that is determined by the false-alarm rate.

The maximum and mean CFAR features are the maximum and mean value of pixels in the CFAR image, respectively. The percent bright CFAR feature is the percentage of pixels within an objected-sized blob that exceed a certain CFAR value. These three contrast-based features are denoted as f_1 , f_2 and f_3 , respectively.

2.2. Semi-supervised Co-training on Lincoln Features

Co-training¹³ generally uses two classifiers based on two sufficient and redundant views, i.e., two feature sets each of which is sufficient for learning classifier and conditionally independent to the other given the class label. In most real-world scenarios, sufficient and redundant views do not exist. Although the sufficiency and redundancy cannot be met, there is also a so called "classification margin" between two classifiers since they uses different features. Classifiers based on different feature sets have different biases, which is an intuitive explanation to why co-training algorithm can succeed [11]. One classifier should know some information that the other classifier does not know. Therefore, the information provided by a classifier should be informative for training the other classifier.

Based on the aforementioned description, we can see that the texture-based features comprehensively depict the texture information of the scatterers. The ranked fill ratio feature reflects the scatterers' characteristic with respect to the power. The contrast-based features mainly concentrate on the difference between the target and its background clutter. They are complementary. Therefore, we propose to apply a texture-based classifier and a contrast-based classifier in a co-training scheme to improve the SAR target discrimination performance. Let X be the input space and $Y = \{0, 1\}$ be the output space. The input space can be partitioned into two views X^1 and X^2 . For any example $x \in X$, we use x^1 and x^2 to denote its two portions under the first view X^1 and the second view X^2 , respectively, i.e., $x = \langle x^1, x^2 \rangle$. In our proposed method, $x^1 = \{\sigma, H_d, \eta\}$ and $x^2 = \{f_1, f_2, f_3\}$. Suppose $L = \{(v_i, y_i) | i = 1, \dots, M, y_i \in Y\}$ contains M labeled training samples and $U = \{u_j | j = 1, \dots, N\}$ contains N unlabeled training samples, where all the labeled and unlabeled samples have two feature sets. In our target discrimination task, we choose the SVM classifier.

The co-training first weakly trains two classifiers using two different feature sets X^1 and X^2 of a small number of labeled samples L , respectively. Each classifier is then used to classified the unlabeled data U . Samples in U with high confident predictions of either classifier are used to augment the training set of the other. This process is repeated until a

certain number of iterations are carried out or there are no unlabeled samples in the training set. The detailed algorithm is described in the following algorithm chart.

Algorithm 1. Procedure of the co-training

Input: a set L^i of labeled training samples on feature set X^i ($i = 1, 2$); a set U of unlabeled training samples.

Output: the optimal SVM classifier F^i on feature set X^i ($i = 1, 2$).

Initialize classifiers F^i under view X^i based on L^i ($i = 1, 2$);

Loop for n iterations:

- 1) Great a pool $U^{i'}$ of samples by choosing u samples at random from $U^{i'}$ and remove these samples from $U^{i'}$ ($i = 1, 2$);
 - 2) Apply classifier F^i to predict labels of unlabeled examples on the portion $u^{i'}$ in $U^{i'}$ ($i = 1, 2$);
 - 3) Choose p most confident positive samples and n most confident negative samples of either view to augment the labeled training set of the other one;
 - 4) Update $F^{i'}$ by learning from the augmented labeled training set $L^{i'}$ ($i = 1, 2$);
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3. EXPERIMENTAL RESULTS

To validate the effectiveness of the semi-supervised target discrimination based on co-training, experiments are performed on the miniSAR data set acquired by Sandia National Laboratories of America [12]. The data set contains nine SAR images whose resolution is $0.1 \times 0.1m$ and size is 1638×2501 . The log-normal-based CFAR is firstly utilized to obtain the chips. Fig.2 shows some chips, including the targets and clutter chips. We take 425 chips from four SAR images as the training set and the remaining 817 chips from other five SAR images as the test set.

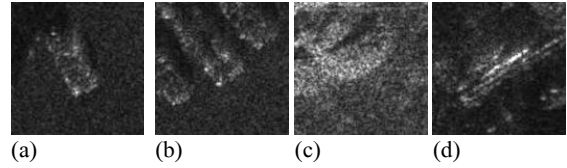


Fig. 2 Some chips after the detection stage. (a) and (b) are target chips. (c) and (d) are clutter chips.

The successfulness of co-training lies in the interactional learning between two different views. In order to demonstrate this, we compare its performance with the self-learning. The self-learning first weakly trains a classifier based on one feature set with a small number of labeled samples and the most predictions on the unlabeled samples are then added into the training data. This process is repeated until the unlabeled samples in the training set are empty. The main difference between the co-training and self-learning is that the former exploits the interactional

learning between two feature sets, while there is no such information exchange in the latter method. We conduct experiments with 16 initial labeled samples, where the number of positive samples is the same as that of negative samples. The initial labeled samples are randomly selected 100 times, and their results are averaged over the 100 different choices. The self-learning uses all the six features. The variations of average accuracy and its standard deviation with the number of iteration number via different methods are shown in Fig. 3. From Fig. 3, we can see that the performance of both self-learning and co-training becomes better and stable with the iteration number increasing. Furthermore, when these two algorithms converge, the average accuracy of co-training is higher than that of self-learning.

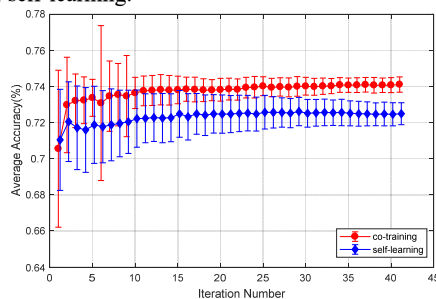


Fig. 3 The variations of average accuracy and standard deviation with the iteration number.

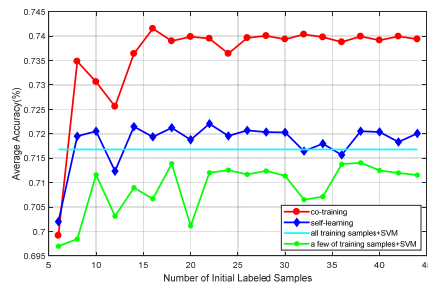


Fig. 4 The variations of average accuracy via different methods with the number of initial labeled samples.

In order to illustrate the performance of our proposed semi-supervised discrimination method, we compare the proposed method with two supervised training schemes and self-learning. Fig. 4 shows the results obtained from different methods. In Fig. 4, “all training samples + SVM” denotes that all the training samples are used to train the SVM; “a few of training samples + SVM” denotes that only a few of training samples, same as those used in our proposed method, are used to train the SVM. Both these two schemes and self-learning use all the six Lincoln features. For “a few of training samples + SVM”, self-learning and our proposed discrimination method, $l(l = 6 \sim 44)$ labeled training samples, where the number of positive samples is

the same as that of negative samples, are randomly selected 100 times, and their results shown in Fig. 4 are averaged over the 100 different choices. In order to make the performance comparison between different methods more clearly, the result of “all training samples + SVM”, which is a constant, is given as a line in Fig. 4. From Fig. 4 we can see that the discrimination performance of our proposed method is better than those of the two supervised schemes and self-learning.

4. CONCLUSIONS

A semi-supervised target discrimination method based on the co-training for SAR images has been presented in this paper to solve the difficulty of obtaining lots of labeled samples. Experimental results on the real SAR images data set demonstrate the effectiveness of the proposed method compared with the conventional supervised discrimination method. Further work will include the better confidence evaluating methods, validations to more data and seeking features appropriate for the co-training algorithm in SAR target recognition.

5. REFERENCES

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